

# Hongxiang (David) Qiu

Assistant Professor  
Department of Epidemiology and Biostatistics  
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## Current position

### Michigan State University

Assistant Professor, Epidemiology and Biostatistics 2023–now  
Research areas: causal inference, survival analysis, semiparametric inference, machine learning

## Education

### University of Washington

PhD, Biostatistics 2016–2021  
Dissertation title: *In pursuit of automated statistical inference under minimal assumptions using machine learning tools*  
Dissertation advisors: Marco Carone, PhD and Alex Luedtke, PhD

### The Chinese University of Hong Kong

B.Sc., Mathematics; Minor in Statistics 2012–2016

## Previous Work Experience

### Wharton School, University of Pennsylvania

Postdoctoral researcher, Statistics 2021–2023  
Supervisor: Eric Tchetgen Tchetgen, PhD and Edgar Dobriban, PhD  
Research areas: machine learning, causal inference, semiparametric inference

### Kaiser Permanente Washington Health Research Institute

Graduate Research Assistant 2016–2021  
PRimary care Opioid Use Disorders Treatment (PROUD) Trial  
Advisor: Jennifer F. Bobb, PhD

## Awards and Fellowships

### University of Washington

Scholarship for 6th Seattle Symposium in Biostatistics October 2020  
2018 Donovan J. Thompson Award October 2018

### Chinese University of Hong Kong

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|--------------------------------------------------------------------------|----------------|
| Dean's List, 2015–2016                                                   | July 2016      |
| College Head's List, 2014–2015                                           | January 2016   |
| Department/Programme Scholarship, 2014–2015                              | January 2016   |
| Undergraduate Mathematics Scheme                                         | November 2015  |
| Shaw College Global Exposure Award Scheme 2014–15                        | April 2015     |
| Heung To Educational Fund Mathematics Scholarship                        | April 2015     |
| Reaching Out Award                                                       | April 2015     |
| 1978 Mathematics Alumnus Li Sze-lim Scholarships                         | January 2015   |
| College Head's List, 2013–2014                                           | January 2015   |
| Department/Programme Scholarship, 2013–2014                              | January 2015   |
| CUHK Golden Jubilee Scholarship for Outgoing Exchange Student, 2014–2015 | September 2015 |
| Yasumoto International Exchange Scholarships                             | July 2014      |
| Dean's List, 2013–2014                                                   | July, 2014     |
| Undergraduate Mathematics Scheme                                         | June 2014      |
| Dragon Crowd "SCHIESSER" Scheme                                          | April 2014     |
| College Head's List, 2012–2013                                           | January 2014   |
| Department/Programme Scholarship, 2012–2013                              | January 2014   |
| Outstanding Essay Award, 2012–2013                                       | November 2013  |
| Dean's List, 2012–2013                                                   | July 2013      |

## Publications

(\*: mentee; †: equal contribution.)

## Methodology

1. **Qiu H** (2024). Non-Plug-In Estimators Could Outperform Plug-In Estimators: a Cautionary Note and a Diagnosis. *Epidemiologic Methods*, 13(1), 1–11. <https://doi.org/10.1515/EM-2024-0008>
2. **Qiu H**, Shi X, Miao W, Dobriban E, Tchetgen Tchetgen E (2024). Doubly robust proximal synthetic controls. *Biometrics*, 80(2). <https://doi.org/10.1093/biomtc/ujae055>.
3. **Qiu, H**, Tchetgen Tchetgen E, Dobriban E (2024). Efficient and multiply robust risk estimation under general forms of dataset shift. *The Annals of Statistics*, 52(4), 1796–1824.
4. **Qiu H**, Cook A, Bobb J (2024). Evaluating tests for cluster-randomized trials with few clusters under generalized linear mixed models with covariate adjustment: a simulation study. *Statistics in Medicine*, 43(2), 201–215. doi: 10.1002/sim.9950.
5. **Qiu H**, Dobriban E, Tchetgen Tchetgen E (2022). Prediction Sets Adaptive to Unknown Covariate Shift. *arXiv preprint: arXiv:2203.06126*. Accepted by *Journal of the Royal Statistical Society, Series B (Statistical Methodology)*.
6. **Qiu H**, Luedtke A (2022). Adversarial Meta-Learning of Gamma-Minimax Estimators That Leverage Prior Knowledge. *Electronic Journal of Statistics*, 17(2), 1996–2043.
7. **Qiu H**, Carone M, Luedtke A (2022). Individualized treatment rules under stochastic treatment cost constraints. *Journal of Causal Inference*, 10(1), 480–493.
8. **Qiu H**, Luedtke A, Carone M (2021). Universal sieve-based strategies for efficient estimation using machine learning tools. *Bernoulli*, 27(4), 2300–2336.

9. **Qiu H**, Carone M, Sadikova E, Petukhova M, Kessler R, Luedtke A (2020). Optimal individualized decision rules using instrumental variable methods. *Journal of the American Statistical Association* (with discussion), 116(533), 174–191.
10. Bobb J, **Qiu H**, Matthews, A, McCormack J, Bradley K (2020). Addressing identification bias in the design and analysis of cluster-randomized pragmatic trials: a case study. *Trials*, 21(1), 289.
11. **Qiu H**, Luedtke A, van der Laan M (2019). Contribution to discussion of “Entropy Learning for Dynamic Treatment Regimes” by Jiang B, Song R, Li J, Zeng D. *Statistica Sinica*, 29(4): 1666–1678.

## Application

12. Al-Sahab B, Kerver JM, Alshaarawy O, Bohnert KM, Elliott MR, **Qiu H**, Jaber A, Neelam H, & Paneth N (2025). Prevalence and characteristics of prenatal cannabis use in Michigan, USA: A statewide population-based pregnancy cohort. *Addiction*. 121(1), 126—137.
13. Wartko PD, Bobb JF, Boudreau DM, Matthews AG, McCormack J, Lee AK, **Qiu H**, Yu O, Hyun N, Idu AE, Campbell CI, Saxon AJ, Liu DS, Altschuler A, Samet JH, Labelle CT, Zare-Mehrjerdi M, Stotts AL, Braciszewski JM, Murphy MT, Dryden D, Arnsten JH, Cunningham CO, Horigian VE, Szapocznik J, Glass JE, Caldeiro RM, Phillips RC, Shea M, Bart G, Schwartz RP, McNeely J, Liebschutz JM, Tsui JJ, Merrill JO, Lapham GT, Addis M, Bradley KA, PROUD Trial Collaborators (2023). Nurse Care Management for Opioid Use Disorder Treatment: The PROUD Cluster Randomized Clinical Trial. *JAMA Internal Medicine*. DOI: 10.1001/jamainternmed.2023.5701.
14. Wartko PD, **Qiu H**, Idu A, Yu O, McCormack J, Matthews AG, Bobb JF, Saxon AJ, Campbell CI, Liu D, Braciszewski JM, Murphy SM, Burganowski RP, Murphy MT, Horigian VE, Hamilton LK, Lee AK, Boudreau DM, Bradley KA (2022). Baseline representativeness of patients in clinics enrolled in the PRimary care Opioid Use Disorders treatment (PROUD) trial: comparison of trial and non-trial clinics in the same health systems. *BMC Health Services Research*, 22(1), 1–11.
15. Fitts W, Tassiou NR, Cisse FA, Vogel A, Atakla HG, Sakadi F, **Qiu H**, Conde ML, Balde AT, Bah AK, Hamani ABD, Anand P, Patenaude B, Mateen F (2019). School Status and its Associations among Children with Epilepsy in the Republic of Guinea. *Epilepsy & Behavior*, 97, 275–281.
16. Jang M, Sakadi F, Tassiou NR, Abass CF, Grundy SJ, Woga A, Kenda BA, Lamine CM, Talibé BA, **Qiu H**, Cohen JM, Carone M, Mateen FJ (2018). Impact of Poorly Controlled Epilepsy in the Republic of Guinea. *Seizure*, 61, 71–77.
17. Zhou Y<sup>†</sup>, **Qiu H**<sup>†</sup>, Xu S (2017). Modeling continuous admixture using admixture-induced linkage disequilibrium. *Scientific Reports*, 7, 1–10.

## Preprint/under review

18. **Qiu H** (2025). Model-Agnostic Bounds for Augmented Inverse Probability Weighted Estimators’ Wald-Confidence Interval Coverage in Randomized Controlled Trials. *arXiv preprint: arXiv:2512.18898*.
19. **Qiu H**, Carone M, Luedtke A, & Gilbert PB. (2025). Multiply Robust Estimation of Conditional Survival Probability with Time-Varying Covariates. *arXiv preprint arXiv:2510.10372*.
20. Si W\*, **Qiu H** (2025). Training-Set Conditionally Valid Prediction Sets with Right-Censored Data. *arxiv preprint: arXiv:2505.02213*.

## Presentations

(\*: mentee; †: equal contribution.)

### Presenters

1. “Berry-Esseen type bound for Structure-Agnostic Semiparametric Treatment Effect Inference in Randomized trials.” **Qiu H.**
  - 2025 Joint Statistical Meeting, Nashville, TN (invited)
  - 2025 American Causal Inference Conference, Detroit, MI (poster)
2. “Novel Estimators of Minimally Important Change for Patient-Reported Outcomes.” **Qiu H.**
  - 2025 Research Day, Department of Epidemiology and Biostatistics, College of Human Medicine, Michigan State University, East Lansing, MI (invited)
3. “Efficient and Multiply Robust Risk Estimation under General Forms of Dataset Shift.” **Qiu H**, Tchetgen Tchetgen E, Dobriban E.
  - 2024 Joint Statistical Meeting, Portland, OR (invited)
4. “Doubly Robust Proximal Synthetic Controls.” **Qiu H**, Xu S, Miao W, Dobriban E, Tchetgen Tchetgen E.
  - 2023 American Causal Inference Conference, Austin, TX
5. “Prediction Sets Adaptive to Unknown Covariate Shift.” **Qiu H**, Dobriban E, Tchetgen Tchetgen E.
  - 2024 5th International Conference on Artificial Intelligence in Electronics Engineering, Bangkok, Thailand (invited)
  - 2022 Joint Statistical Meeting, Washington, D.C.
  - 2022 American Causal Inference Conference, Berkeley, CA (poster)
6. “Optimal individualized decision rules using instrumental variable methods.” **Qiu H**, Carone M, Sadikova E, Petukhova M, Kessler R, Luedtke A.
  - 2021 Joint Statistical Meetings, virtual (invited)
  - 2020 ENAR Spring Meeting, virtual
7. “Constructing asymptotically normal plug-in estimators with highly adaptive Lasso and data adaptive series.” **Qiu H**, Luedtke A & Carone M.
  - 2019 WNAR/IMS/JR Annual Meeting, Portland, OR

### Contribution to presentations

8. “Robust Estimation of Stochastic Intervention Effects under Multistage Missingness.” Song S\*, **Qiu H**, Chen H, Parks C & Luo Z. American Causal Inference Conference, Salt Lake City, UT, 2026
9. “Prevalence and characteristics of prenatal cannabis use in Michigan: Results from a statewide population-based pregnancy cohort.” Al-Sahab B, Alshaarawy O, Bohnert K, Jaber A, **Qiu H**, Kerver J & Paneth N. 37th Annual Meeting of the Society for Pediatric and Perinatal Epidemiologic Research (SPER), Austin, TX, 2024.

10. “Prevalence and characteristics of prenatal cannabis use in Michigan: Results from a statewide population-based pregnancy cohort.” Al-Sahab B, Alshaarawy O, Bohnert K, Jaber A, **Qiu H**, Kerver J & Paneth N. 37th Annual Meeting Society for Pediatric and Perinatal Epidemiologic Research, Austin, TX, 2024.
11. “Statistical challenges in the design of a pragmatic trial of primary care-based treatment for opioid use disorders: the PROUD trial.” Bobb JF, **Qiu H**, Matthews A & Bradley K. 12th International Conference on Health Policy Statistics, Charleston, SC, 2018

## Reviewer for Funding Agencies

- 2026: Patient-Centered Outcomes Research Institute (PCORI)
- 2025: National Institutes of Health (NIH)
- 2024: UK Research and Innovation

## Reviewer for International Journals and Conferences

- *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*
- *Journal of the American Statistical Association*
- *Biometrics*
- *Journal of Causal Inference*
- *Biostatistics*
- *Statistica Sinica*
- *Journal of Computational and Graphical Statistics*
- *Electronic Journal of Statistics*
- *International Journal of Biostatistics*
- *Statistical Methods in Medical Research*
- *Stat*
- *Statistical Science*
- 5th International Conference on Artificial Intelligence in Electronics Engineering (AIEE 2024)
- 37th Conference on Uncertainty in Artificial Intelligence (UAI 2021)

## Mentoring Experience

|                                                                                    |          |
|------------------------------------------------------------------------------------|----------|
| Gu Gong                                                                            | 2024–now |
| PhD student, Department of Epidemiology & Biostatistics, Michigan State University |          |
| Shengfang Song                                                                     | 2024–now |

PhD student, Department of Epidemiology & Biostatistics, Michigan State University

Xi Gong 2024–now  
Department of Statistics & PhD student, Probability, Michigan State University

Wenwen Si 2024–2026  
Department of Computer & PhD student, Information Science, University of Pennsylvania

## Teaching Experience

### Michigan State University

EPI-826: Research Methods in Epidemiology 2025–now  
*Instructor*

Topics including: GLM, GEE, GLMM, sandwich robust SE, bootstrap, power calculation

EPI-920: Inference under Semiparametric and Nonparametric Models (Independent Study) Spring 2024  
*Instructor*

Topics including: (efficient) influence function, semiparametric efficiency bound, one-step correction, multiple robustness

EPI-826: Research Methods in Epidemiology Fall 2024  
*Guest instructor*

Topics including: generalized linear mixed model, generalized estimating equation

### University of Washington

STAT 583: Advanced Theory of Statistical Inference Spring 2019  
*Tutor*

Topics including: empirical processes, semiparametric efficiency

Instructors: Alex Luedtke, PhD and Marco Carone, PhD

BIOST 557: Applied Statistics and Experimental Design Winter 2019  
*Teaching Assistant*

Topics including: one/two-sample t-test, linear models, GLMs, causation versus correlation  
Instructor: Brian Lerous, PhD

STAT 582: Advanced Theory of Statistical Inference Winter 2018  
*Grader*

Topics including: Bayes methods, decision theory, UMPU test  
Instructor: Jon A. Wellner, PhD

## Software

|                             |                                                                                                                                                                     |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CanonicalFamilyExtra</b> | Extra canonical link family objects for generalized linear models. Particularly useful for fitting logistic or log-linear models for unbounded continuous outcomes. |
| <b>RiskEstDShift</b>        | Efficient risk estimation under dataset shift.                                                                                                                      |
| <b>APACpredset</b>          | Asymptotically Probably Approximately Correct prediction sets under unknown covariate shift.                                                                        |
| <b>CAMer</b>                | Continuous Admixture Modeling based on the result of iMAAPs.                                                                                                        |